COLM 2026

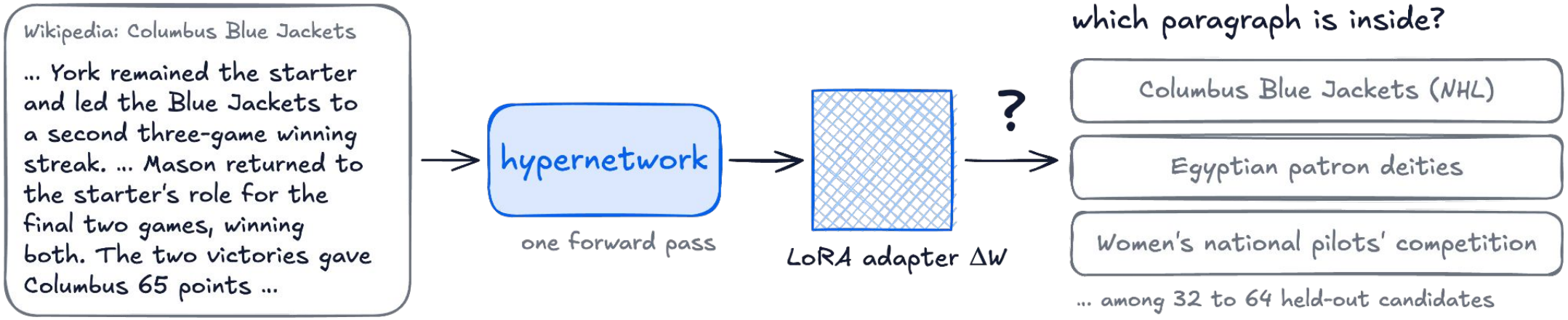
Reading Documents out of Weight Updates: Hypernetwork-Written LoRA Adapters Linearly Encode Their Document

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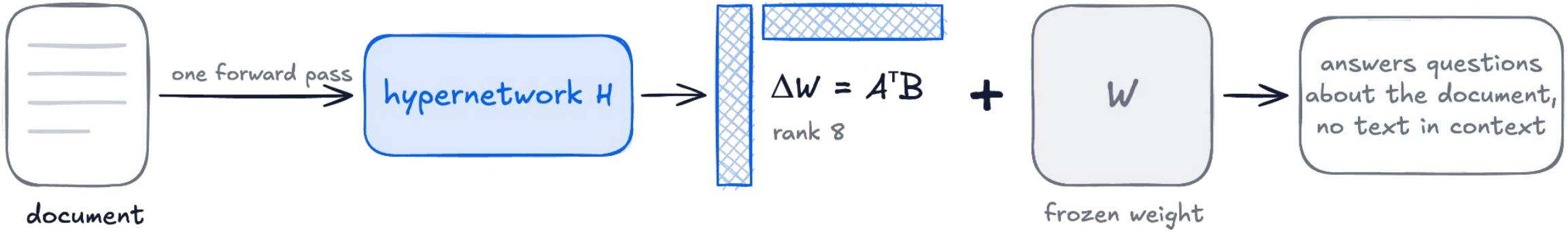
Context Beyond the Window Workshop, COLM 2026

Motivation

- **Auditing:** was my copyrighted document written into this adapter?
- **Privacy**: can an attacker read a document with my private info back out of an adapter?
- **Reconstruction** works from text embeddings [Morris et al, 2023], and it is rapidly evolving for weight updates



Background: Hypernetworks and Doc-to-LoRA



- **Hypernetwork** is a neural network that outputs the weights for another neural network [Ha et al., 2017]
- **Doc-to-LoRA** [Charakorn et al., 2026] is a hypernetwork that converts a raw document into ΔW in one forward pass. Adding that ΔW to a base model gives it context.
- They **released** hypernetwork checkpoints for three base models, which we use

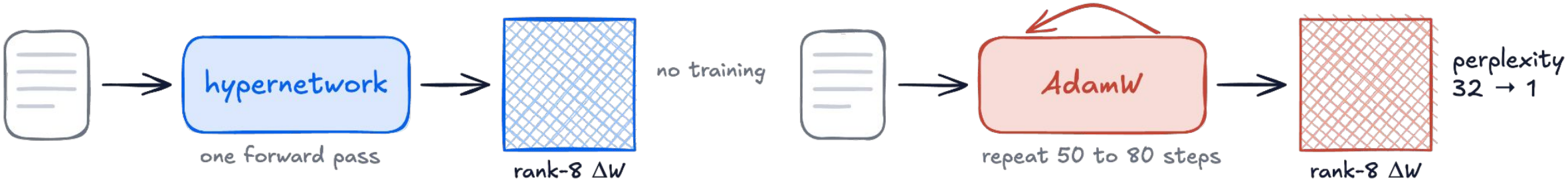
We study 2 ways of writing into weights

Hypernetwork (Doc-to-LoRA)

- One forward pass per document (instant)
- No training per document

AdamW fine-tuning

- A fresh rank-8 adapter per document
- 50 to 80 training steps



- We use the same factor shape for both, rank 8 on the MLP down_proj
- AdamW adapters learn their document since Gemma's perplexity goes from 32 to 1

Questions we aim to answer

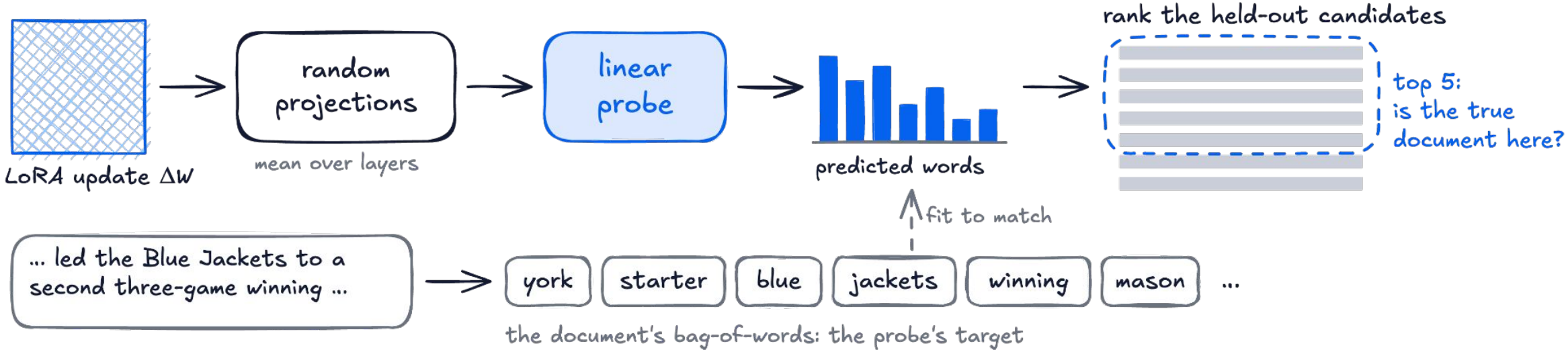
1. Can a linear probe identify the document from an adapter's weights? Does that depend on how the adapter was written?

2. What makes a hypernetwork-written update readable if an AdamW update of the same document is not?

3. How can you identify the document that a publicly released adapter was trained on? Can somebody use noise to hide it?

Identification = given an adapter and N candidate documents, find the one it was trained on

Probing the weight update

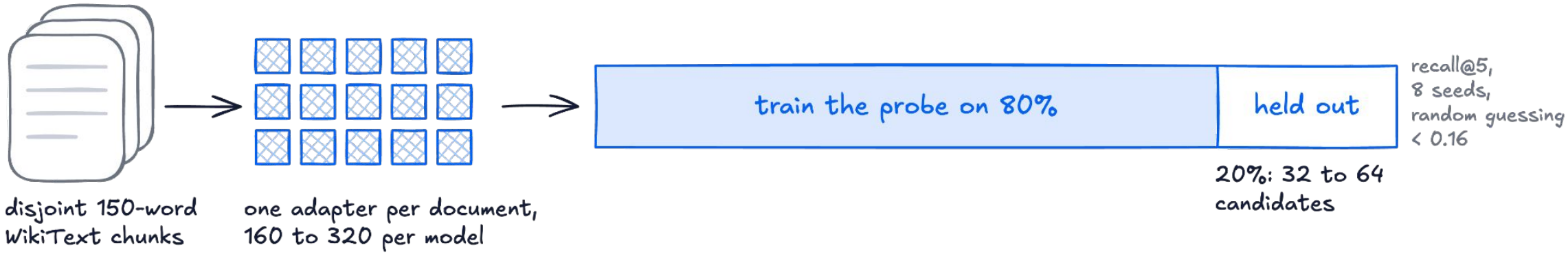


- **Goal** = identify the document among candidates
- **Recall@5** = share of held-out adapters whose true document ranks in the top 5 candidates
- A linear probe asks whether the document is part of the weight update, borrowing from Alain & Bengio, 2017

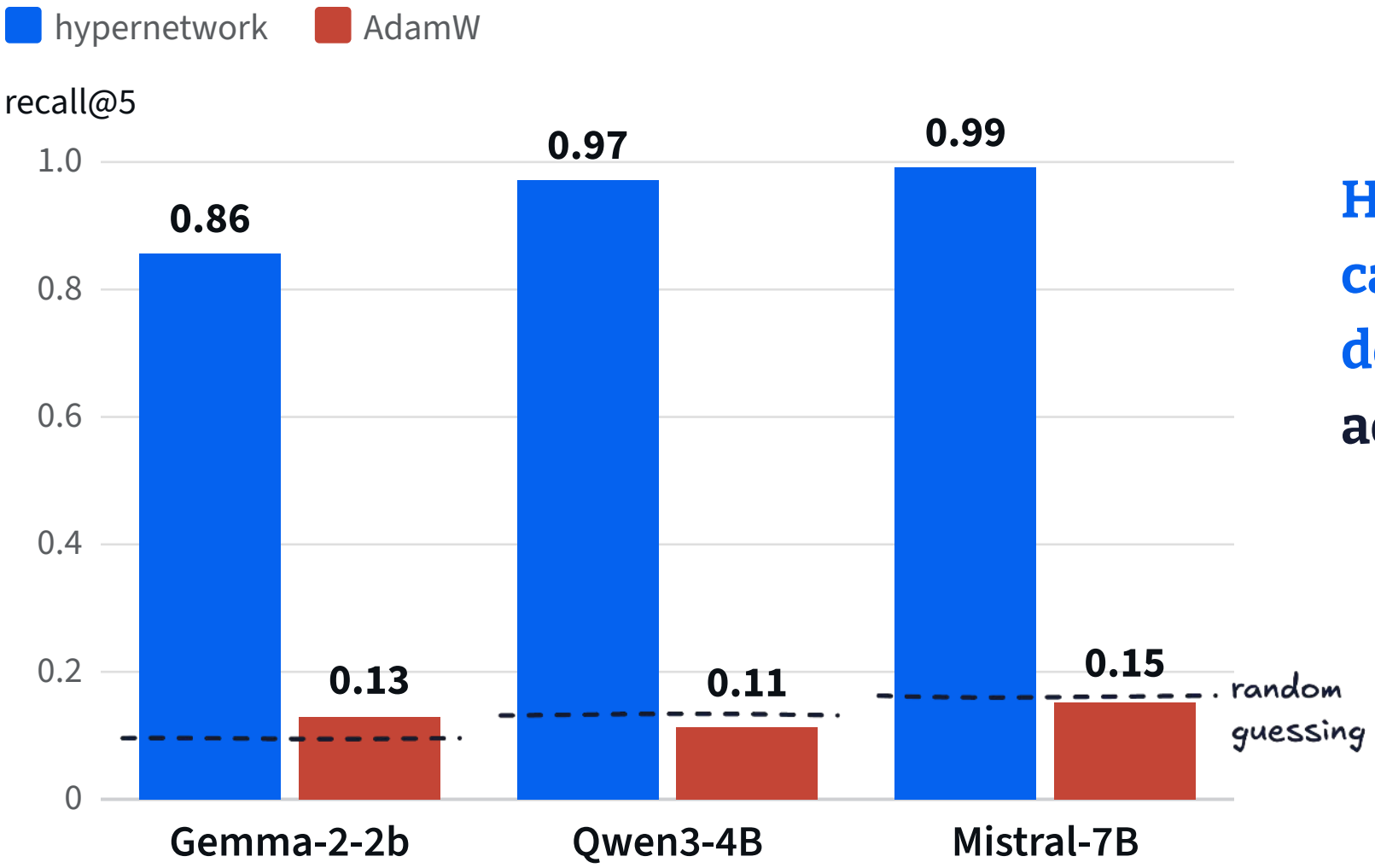
Experimental Setup

- **Base models:** Gemma-2-2b-it, Qwen3-4B-Instruct, Mistral-7B-Instruct v0.2
- **Documents:** disjoint 150-word WikiText chunks
- **Adapters:** 160 to 320 per model, one per document
- **Metric:** recall@5 among 32 to 64 held-out candidates

more details in the paper



Results · RQ1

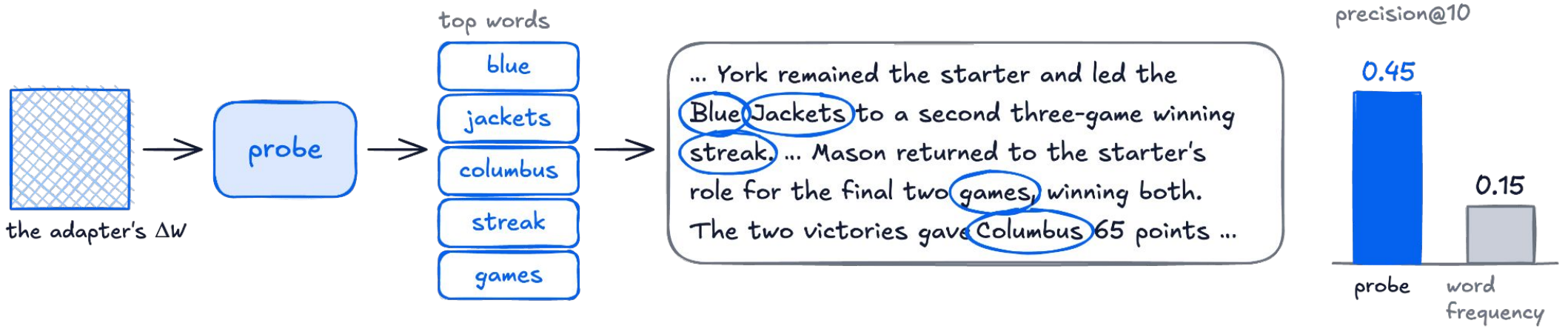


Hypernetwork adapters can recall their training doc, while AdamW adapters can't

RQ1 · Recovered words

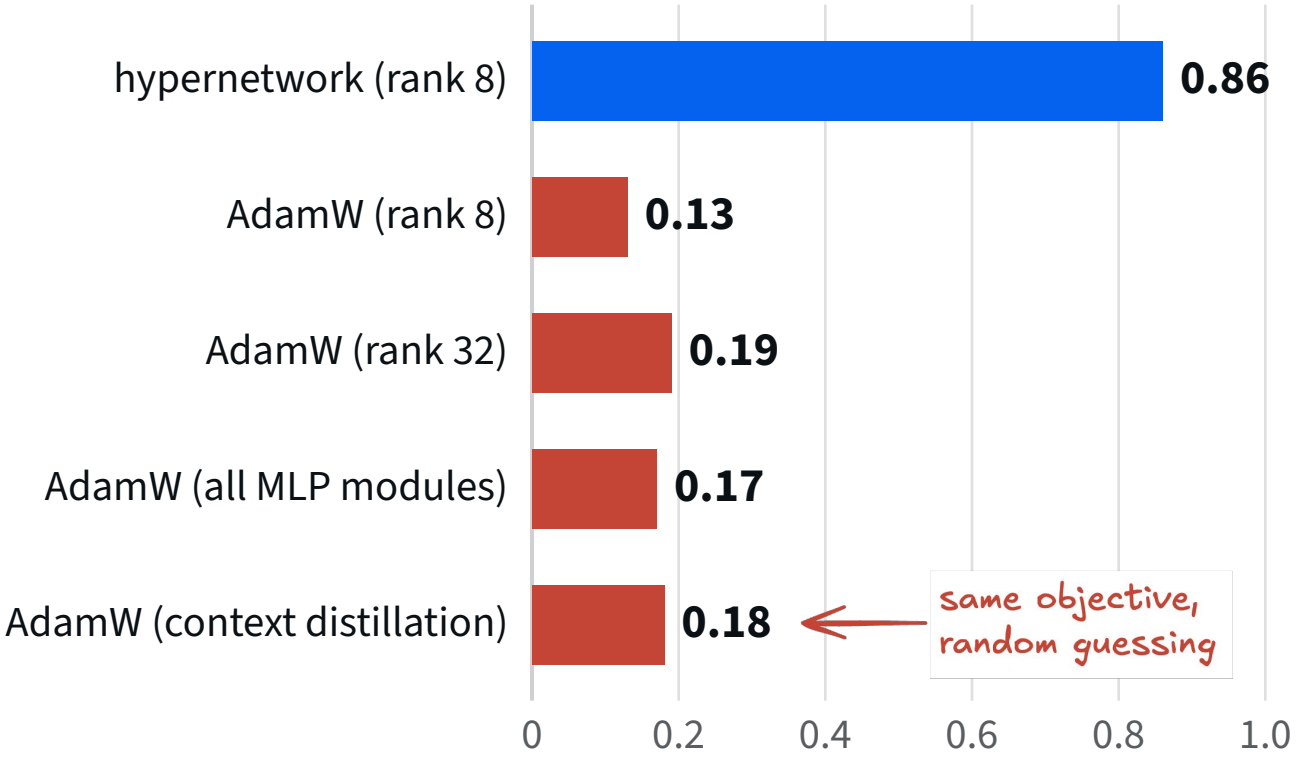
Held-out document (topic)	Words recovered from ΔW (without text access)
Columbus Blue Jackets (NHL)	**blue, jackets, columbus, streak, games**
Egyptian patron deities	**deity, kingdom, amun, name**

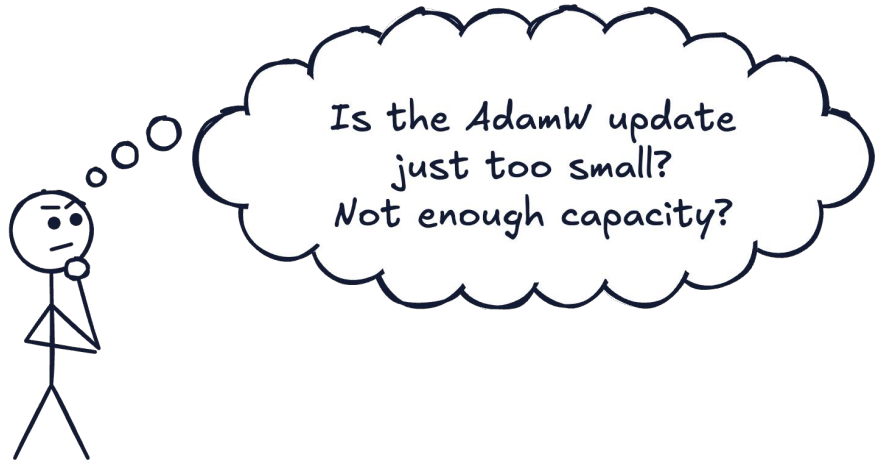
- **Precision@10:** 0.45 for our probe vs. 0.15 for a word-frequency baseline



RQ1 · Controls

Recall@5 on Gemma-2-2b





- **Update norm:** 6.7 (hyper-network) vs. 5.9 (AdamW)
- Readability depends on how the update was written, not on its size, capacity or training objective.

RQ1 · Does it hold for other hypernetworks?

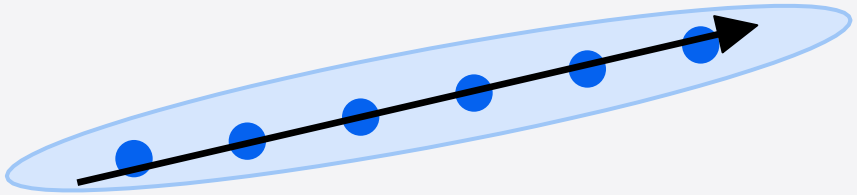
Generator (Gemma-2-2b)	target	recall@5	AdamW
Doc-to-LoRA (released)	MLP down_proj	**0.86**	**0.13**
Our hypernetwork (trained from scratch)	MLP down_proj	**0.76**	**0.12**
Text-to-LoRA [Charakorn et al., ICML 2025]	attention q_proj	**0.85**	**n/a**

All three hypernetworks are decodable, but none of the AdamW adapters we tried are.

Results · RQ2 · Why is the update readable?

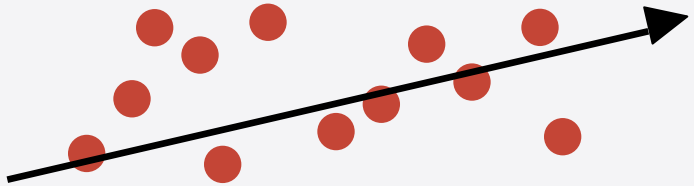
- We believe that a hypernetwork is a smooth map from a document embedding to the update

Hypernetwork



updates lie on a low-dimensional subspace

AdamW



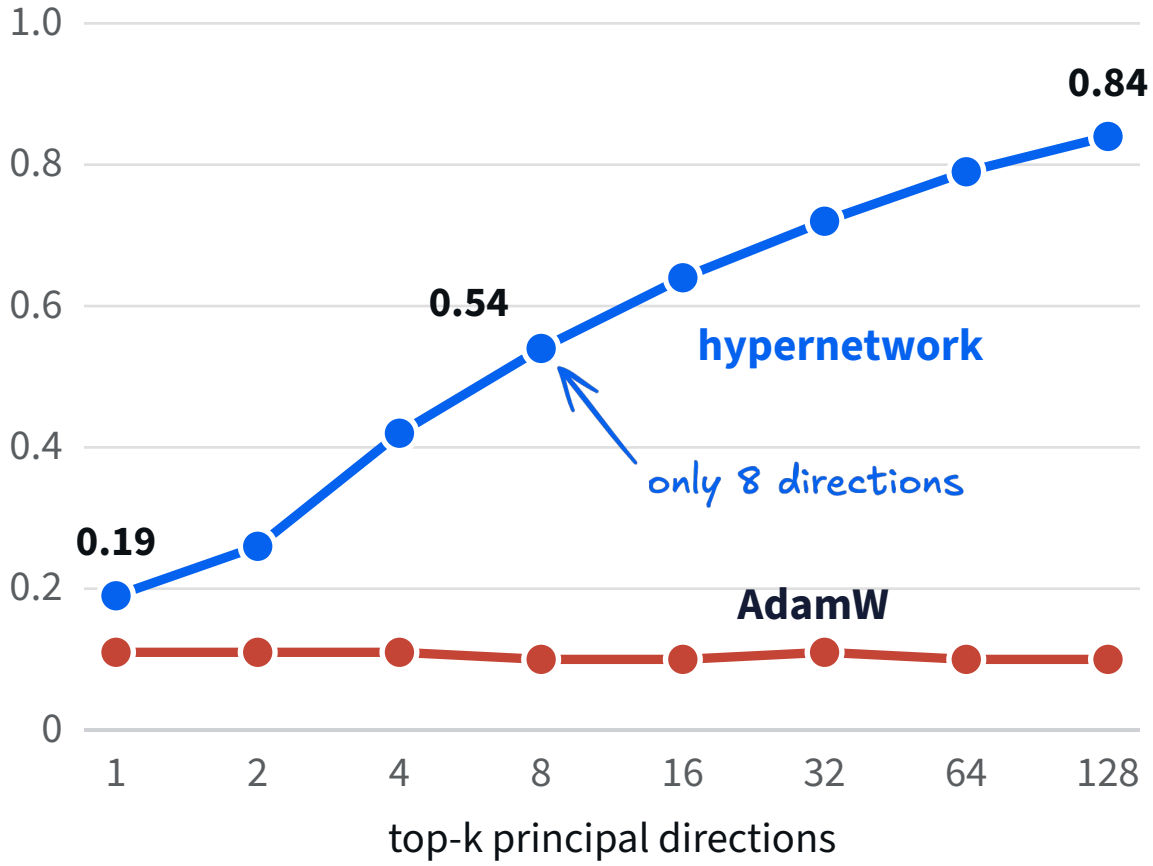
updates fill many directions

- **Linear fit from embedding to update (R^2 on held-out adapters)**

  Our hypernetwork (trained from scratch) = 0.97, while AdamW < 0

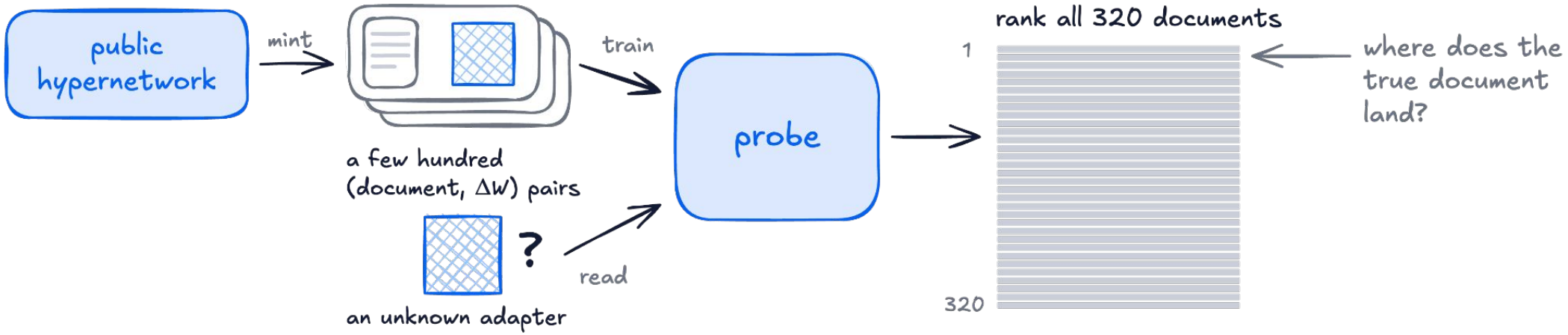
RQ2 · A low-dimensional subspace

Recall@5 using only the top-k principal directions (Gemma-2-2b)



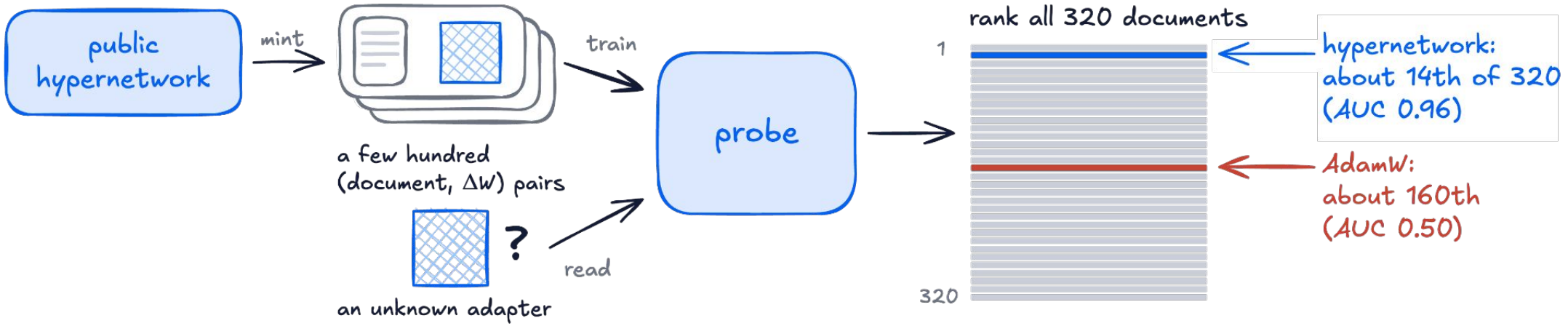
- **Top 8 directions:** 0.54 (hypernetwork) vs. 0.10 (AdamW)

Results · RQ3 · Real-world application



- **Threat model:** we have the released adapter, the public hypernetwork and 320 candidates
- **Method:** mint (document, ΔW) pairs with the hypernetwork and train the probe on them
- **Metric:** AUC = chance the true document ranks above a random wrong one

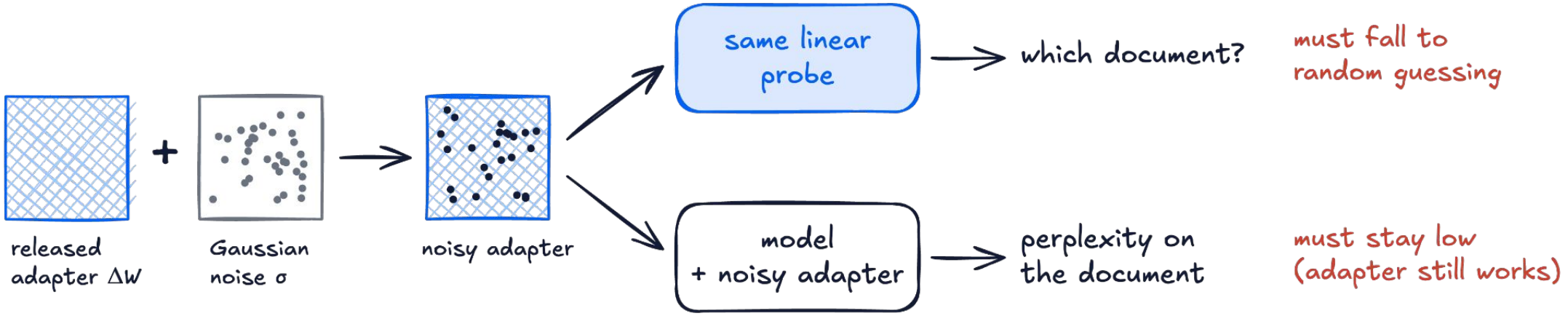
RQ3 · Was my document written into this adapter?

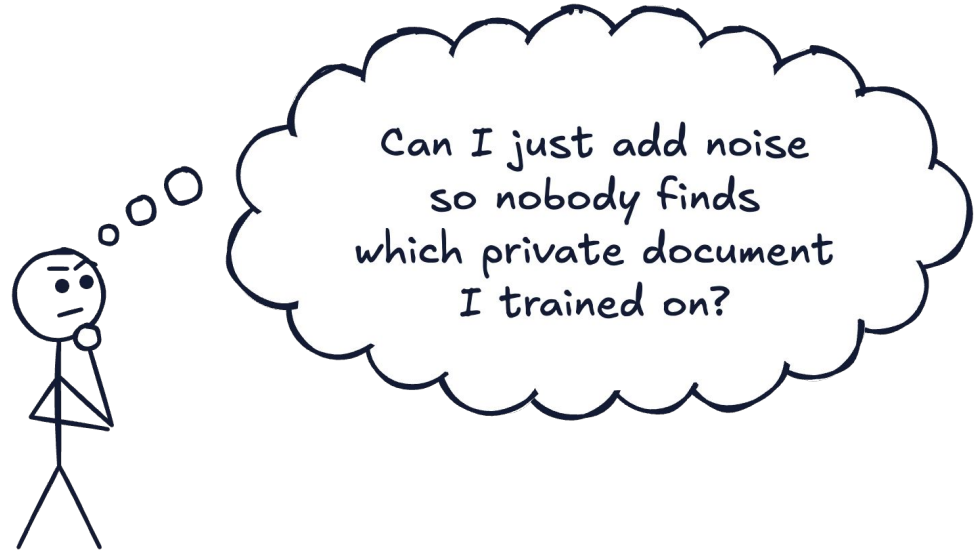


- **Why AUC:** with 1200 documents, recall@5 falls to 0.32, but AUC stays at 0.98
- **Privacy:** the probe's top 20 words recover a third of the document's named entities and numbers (0.32 vs. 0.10 for word frequency)

Anyone with the public hypernetwork can check which document an adapter holds.

RQ3 · Can somebody use noise to hide a document?

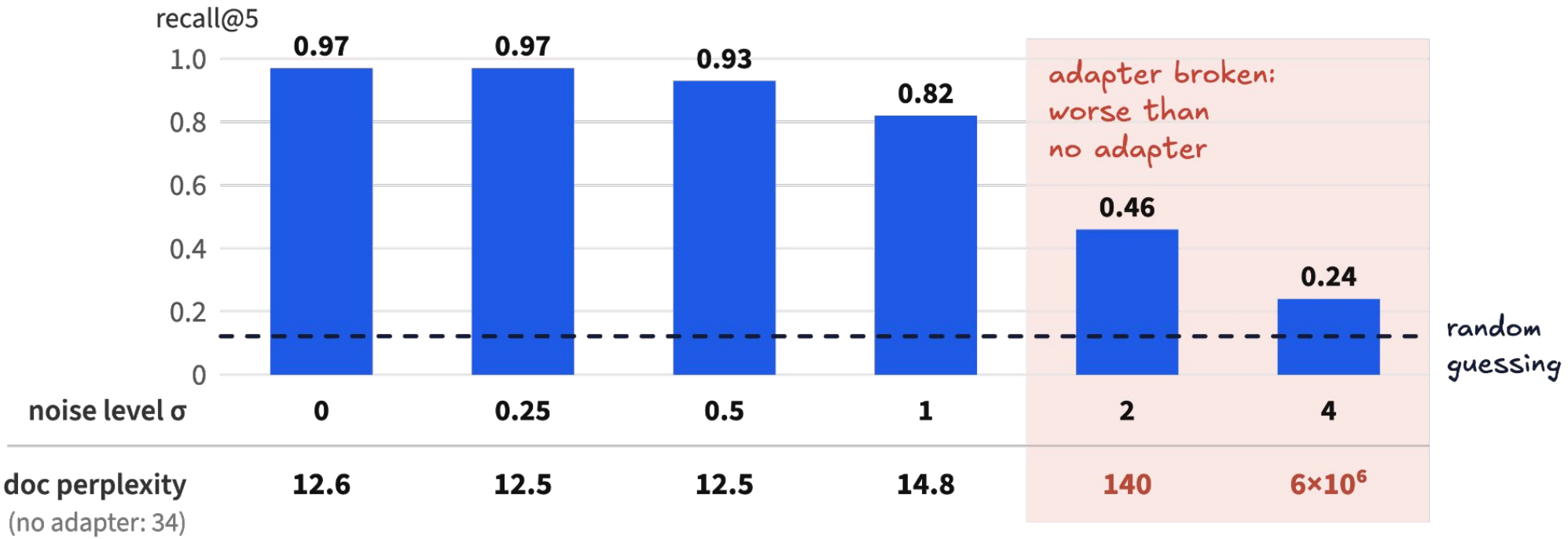




- **Method:** add Gaussian noise to the adapter's weights before releasing it
- **Target:** the probe can no longer identify the doc, but the adapter still works

RQ3 · Can somebody use noise to hide a document?

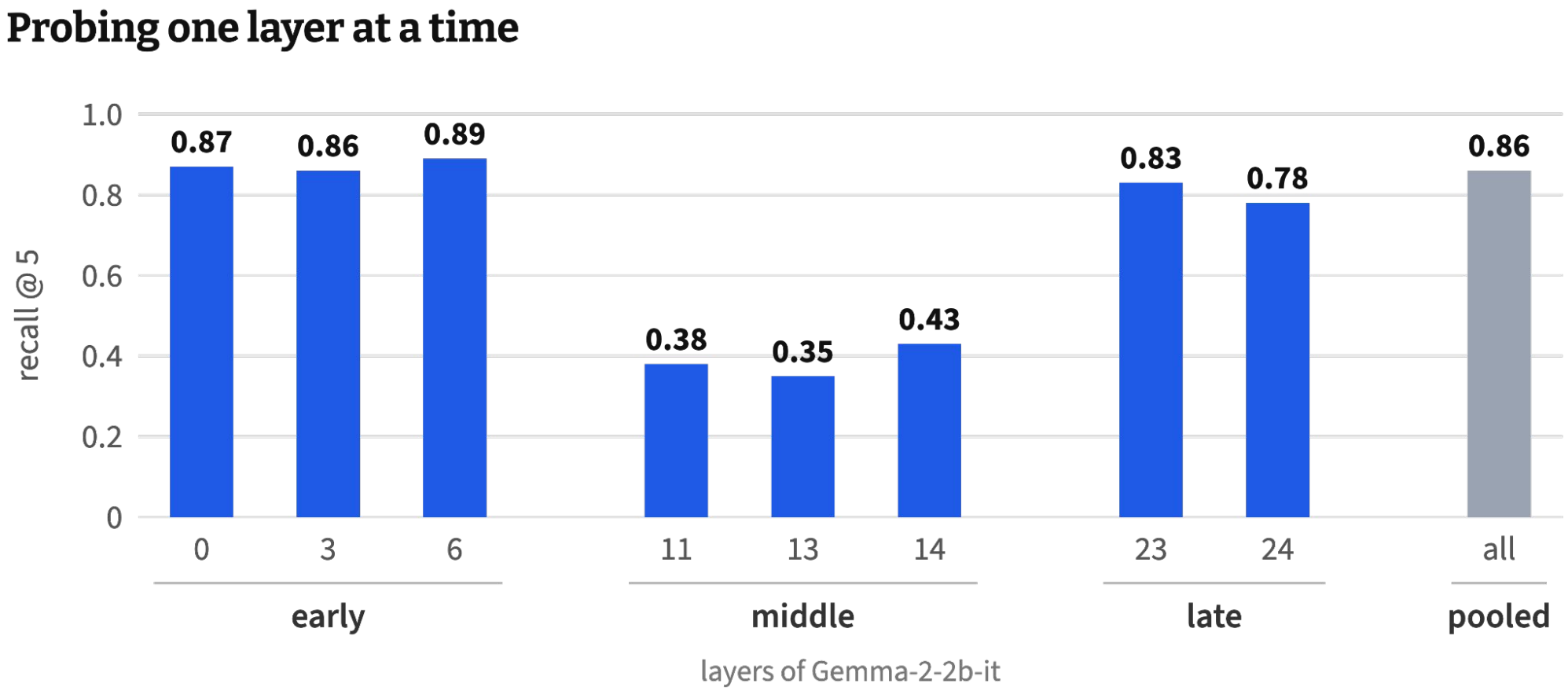
Adding Gaussian noise to the adapter's weights (Gemma-2-2b)



σ is relative to each weight matrix's standard deviation

You can't use noise to hide the document and still expect the adapter to work

Extra · Which layers recall the document?



One early layer recalls the document about as well as all the layers pooled.

Summary

1. Can a linear probe identify the document from an adapter's weights? Does that depend on how the adapter was written?

2. What makes a hypernetwork-written update readable if an AdamW update of the same document is not?

3. How can you identify the document that a publicly released adapter was trained on? Can somebody use noise to hide it?

Summary

1 Can a linear probe identify the document from an adapter's weights?

Yes, if a hypernetwork wrote it, on all 3 base models

2 What makes a hypernetwork-written update readable if an AdamW…?

For our hypernetwork, we find that the update is close to a linear function of the document embedding.

3 How can you identify the document? Can somebody use noise to hide it?

Train the probe on pairs from the public hypernetwork (AUC 0.96).

No, noise can't hide it without breaking the adapter.

Future directions

- Find a defense that hides the document without breaking the adapter
- Reconstruct the document's text from the adapter, not just identify it
- Develop more privacy tests to check for training data identification
- Build an index of adapters, to fetch the one that holds a document
- Effect of release-time changes like lower rank, quantization

Thank you!

TL;DR

Hypernetwork-written LoRA adapters linearly encode their document, so a linear probe on the weight update recalls it while AdamW adapters can't. Noise can't hide it without breaking the adapter.



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